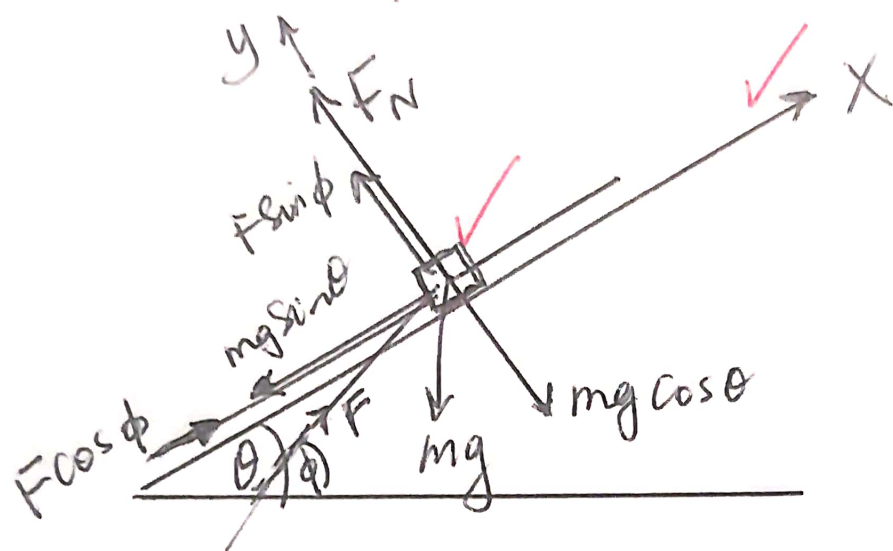


EMO: Sick/Agrotat and Supplementary Exams



[11]

$$\phi = 15.0^\circ$$

$$\theta = 25.0^\circ$$

$$F_x = 0$$

$$\cos \phi - mg \sin \theta + f_s = 0$$

$$+ \mu_s mg \cos \theta - \mu_s F \sin \phi$$

$$= mg \sin \theta$$

$$\sum F_y = 0$$

$$F_N + F \sin \phi = mg \cos \theta$$

$$\therefore F_N = mg \cos \theta - F \sin \phi$$

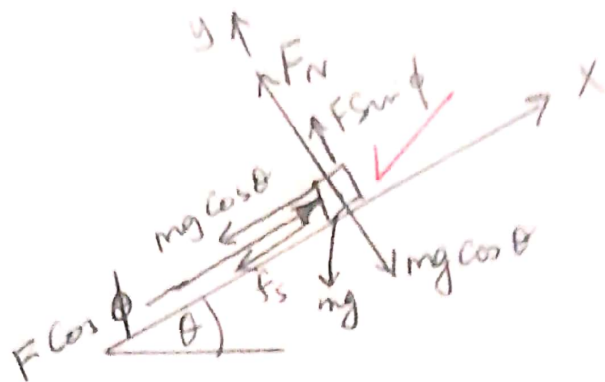
$$\cos \phi - \mu_s \sin \phi = mg \sin \theta - \mu_s mg \cos \theta$$

$$\therefore F = \frac{mg (\sin \theta - \mu_s \cos \theta)}{(\cos \phi - \mu_s \sin \phi)}$$

$$= \frac{90.0 \text{ N} (\sin 25.0^\circ - 0.370 \cos 25.0^\circ)}{(\cos 15.0^\circ - 0.370 \sin 15.0^\circ)}$$

$$= 9.03 \text{ N}$$

(5)



$$\Sigma F_x = 0$$

$$F \cos \phi - mg \sin \theta - f_s = 0$$

$$F \cos \phi - \mu_s F_N = mg \sin \theta$$

$$F \cos \phi - \mu_s mg \cos \theta + F \mu_s \sin \phi = mg \sin \theta$$

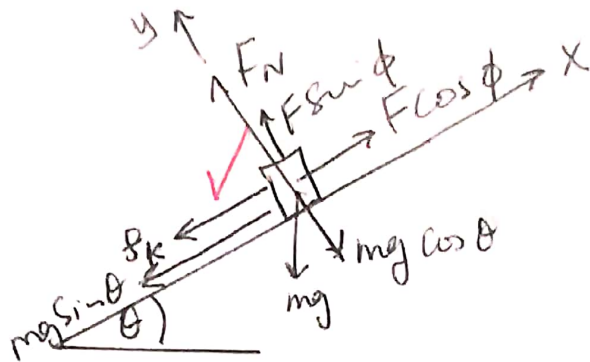
$$F \cos \phi + \mu_s F \sin \phi = mg \sin \theta + \mu_s mg \cos \theta$$

$$F (\cos \phi + \mu_s \sin \phi) = mg (\sin \theta + \mu_s \cos \theta)$$

$$\therefore F = \frac{mg (\sin \theta + \mu_s \cos \theta)}{(\cos \phi + \mu_s \sin \phi)} \checkmark$$

$$= \frac{90.0 \text{ N} (\sin 25.0^\circ + 0.375 \cos 25.0^\circ)}{(\cos 15.0^\circ + 0.375 \sin 15.0^\circ)}$$

$$= 64.3 \text{ N} \checkmark$$



$$\Sigma F_x = ma = 0 \text{ for constant velocity}$$

$$F \cos \phi - mg \sin \theta - f_k = 0$$

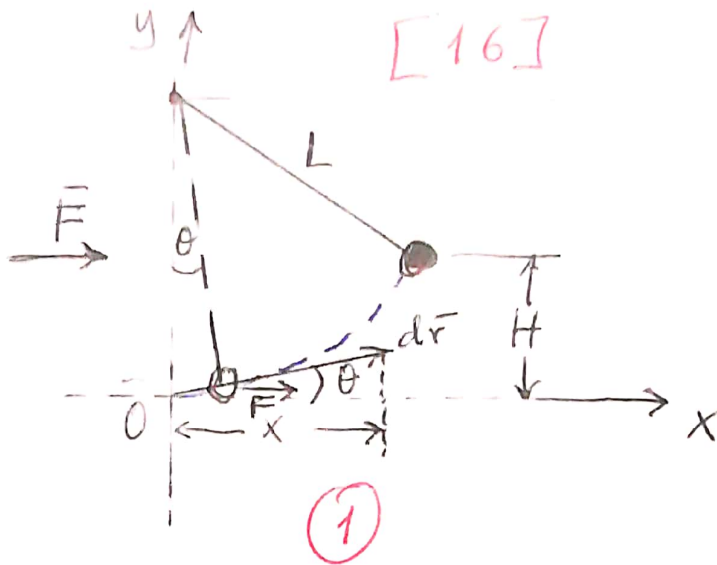
$$F \cos \phi - f_k = mg \sin \theta$$

$$F \cos \phi - \mu_k (mg \cos \theta - F \sin \phi) = mg \sin \theta$$

$$\therefore F = \frac{mg (\sin \theta + \mu_k \cos \theta)}{(\cos \phi + \mu_k \sin \phi)} \quad \checkmark \quad (3)$$

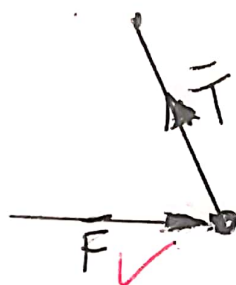
$$= \frac{90.0 \text{ N} (\sin 25.0^\circ + 0.156 \cos 25.0^\circ)}{(\cos 15.0^\circ + 0.156 \sin 15.0^\circ)}$$

$$= 50.4 \text{ N} \quad \checkmark$$



(a) Non-isolated ✓

(b)



Tension, $\vec{T}$
Wind, $\vec{F}$

(2)

(c) $W_{\text{other}} - f_k d = \Delta K + \Delta U$

$\Delta K + \Delta U_g = W_{\text{ext}} = W_T + W_F$

(3)

(d) (i) $W_T = \int_{r_{\text{initial}}}^{r_{\text{final}}} \vec{T} \cdot d\vec{r}$ and $W_F = \int_{r_{\text{initial}}}^{r_{\text{final}}} \vec{F} \cdot d\vec{r}$

(2)

(ii) $\int \vec{T} \cdot d\vec{r} = 0$ since $\vec{T}$ is perpendicular to $d\vec{r}$ during path of ball.

(e) $\Delta K + \Delta U_g = W_F$

The ball has zero speed at the initial and at the final positions $\Rightarrow \Delta K = 0$.

Also, $\vec{F}$ acts in the direction of the x-component of $d\vec{r}$ since we are taking a dot product $\vec{F} \cdot d\vec{r}$

$\therefore \Delta U_g = |\vec{F}| |d\vec{r}| \cos \theta$

$mg y_f - mg y_i = F x$

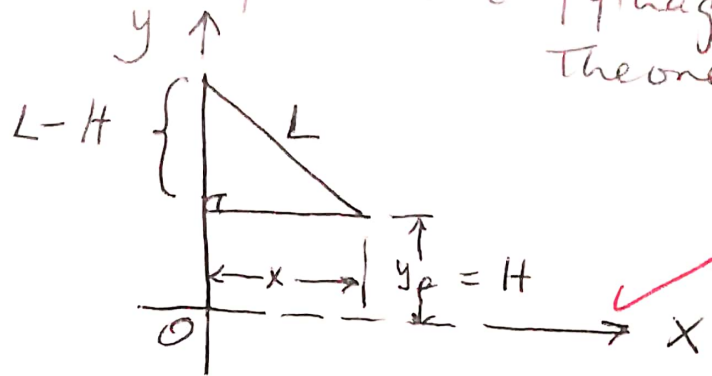
$mg y_f = F x$ after letting $y_i = 0$ in our co-ordinate axes.

At the final position of the ball, $y_f = H$

$$\Rightarrow mgH = F_x$$

$$\Rightarrow H = \frac{F_x}{mg} \checkmark$$

(f) $x^2 + (L-H)^2 = L^2$ from the Pythagorean Theorem.



$$\therefore x^2 = L^2 - (L-H)^2$$
$$= L^2 - (L^2 - 2LH + H^2)$$

$$\therefore x = \sqrt{2LH - H^2} \checkmark$$

(g) $H = \frac{F_x}{mg}$

$$= \frac{F \sqrt{2LH - H^2}}{(mg)}$$

$$\Rightarrow H^2 = \frac{F^2 (2LH - H^2)}{(mg)^2}$$

$$(mg)^2 H^2 = 2LHF^2 - H^2 F^2$$

$$(mg)^2 = \frac{2LF^2 - F^2}{H}$$

$$H[(mg)^2 + F^2] = 2LF^2$$

$$\therefore H = \frac{2LF^2}{(mg)^2 + F^2} = \frac{2L}{\left(\frac{mg}{F}\right)^2 + 1} \checkmark$$



Assume isolated system with no leaks.

$$\therefore -Q_{\text{hot}} = Q_{\text{cold}}$$

$$-\left[Q_{\text{phase A}} + Q_{\text{lost by hot steam to reach } T_f} \right] = Q_{\text{phase A}} + Q_{\text{raise melted water temp from } 0^\circ\text{C} \rightarrow T_f}$$

$$-\left[-L_v m_s + m_s c_w (T_f - 100^\circ\text{C}) \right] = L_f m_i + m_i c_w (T_f - 0^\circ\text{C})$$

$$L_v m_s - m_s c_w (T_f - 100^\circ\text{C}) = L_f m_i + m_i c_w (T_f - T_i) \quad \text{when } T_i = 0^\circ\text{C} \quad \checkmark$$

$$L_v m_s - m_s c_w T_f + m_s c_w (100^\circ\text{C}) = L_f m_i + m_i c_w T_f$$

$$L_v m_s + m_s c_w (100^\circ\text{C}) - L_f m_i = m_i c_w T_f + m_s c_w T_f$$

$$\therefore T_f = \frac{L_v m_s + m_s c_w (100^\circ\text{C}) - L_f m_i}{c_w (m_i + m_s)} \quad \checkmark$$

Parameters looked up; and given ones:

$$c_s = 2010 \text{ J/kg} \cdot ^\circ\text{C} @ 5^\circ\text{C} \text{ is ok!}$$

$$L_{v,s} = 2.26 \times 10^6 \text{ J/kg}$$

$$c_w = 4186 \text{ J/kg} \cdot ^\circ\text{C}$$

$$L_{f,ia} = 3.33 \times 10^5 \text{ J/kg}$$

$$m_s = 14.0 \text{ g} = 0.0140 \text{ kg}$$

$$T_{i,s} = 100^\circ\text{C}$$

$$m_{ice} = 47.0 \text{ g} = 0.0470 \text{ kg}$$

$$T_{i,ia} = 0^\circ\text{C}$$

Substitution:

$$T_f = \frac{\left[(2.26 \times 10^6 \text{ J/kg})(0.014 \text{ kg}) + (0.014 \text{ kg})(4186 \text{ J/kg} \cdot ^\circ\text{C})(100^\circ\text{C}) - (3.33 \times 10^5 \text{ J/kg} \cdot ^\circ\text{C})(0.047 \text{ kg}) \right]}{(4186 \text{ J/kg} \cdot ^\circ\text{C})(0.047 \text{ kg} + 0.014 \text{ kg})}$$

$$(a) \quad (4186 \text{ J/kg} \cdot ^\circ\text{C})(0.047 \text{ kg} + 0.014 \text{ kg})$$

$$= 21849.4 \text{ J} / 255.346 \text{ J/}^\circ\text{C} = 85.6^\circ\text{C} \quad \checkmark$$

(7)

From the energy released by the phase change of steam to water alone, there is enough to melt all of the 47.0 g ice.

It is possible that the problem has been broken into subsections...

(b) For the new mass of steam, we need to re-evaluate the heat available on our LHS of the energy balance equation and then apportion it to first phase changing the ice and the remainder to raising the water temperature.

From equation (A); the heat available is:

$$L_v m'_s - m'_s c_w T_f + m'_s c_w (100^\circ\text{C}) = Q_{\text{total avail.}} T_f \text{ is now } 0^\circ\text{C}$$

$$Q_{\text{total avail.}} = (2.26 \times 10^6 \text{ J/kg})(0.0014 \text{ kg}) - 0 + (0.0014 \text{ kg})(4186 \text{ J/kg}\cdot^\circ\text{C})(100^\circ\text{C})$$
$$= 3750.04 \text{ J} \quad (4)$$

We are now in a position to determine the amount of ice at 0°C that could be melted by this available heat energy.

$$\Rightarrow Q_{\text{total avail.}} = L_F m'_{\text{ice}} \quad \checkmark$$

$$\Rightarrow m'_{\text{ice}} = \frac{Q_{\text{total avail.}}}{L_F} = \frac{3750.04 \text{ J}}{3.33 \times 10^5 \text{ J/kg}} = 11.3 \text{ g}$$

The ice that will be melted = 11.3 g , and the rest will still be ice at 0°C . The melted ice will also remain at 0°C .

Only one bonus point could be awarded for very good logical explanations that serve to link up "sub-sections" of the problem.